

VBF-T9R4-DSH-001 Datasheet (Target vs Measured)

t9_r4 - grid-storage cell cathode on the Chen2020 NMC811/graphite profile

Result: HONEST NEGATIVE (no candidate satisfies the full contract; nothing reported as passing)

| Criterion | Threshold | Best measured | Verdict |

|---|---|---|---|

| Computed avg voltage | ≥ 4.6 V | 4.6413 V (AIB55) | window pass, then REJECTED on compatibility |

| Charging potential | ≤ 4.8 V | 7.410 V (AIB55, $V(0.4 \rightarrow 0.3)$) | FAIL - decisive |

| Catalogue exclusion | not of 115 | PASS (outside envelope) | pass |

| Hull exclusion | outside 102-point hull | PASS (outside envelope) | pass |

| Supported family | six families | layered LiMO2 | pass |

| Energy density 1C | ≥ 327.18 Wh/kg | not reached (no survivor to stage 3) | n/a |

| ED_active report | $V \times C \times 0.9$ | AIB55: $4.6413 \times 324.38 \times 0.9 = 1355$ (Wh/kg-active, informative only) | n/a |

| 4C/45C plating | none | not reached | n/a |

| Thermal runaway | no trigger | not reached | n/a |

| SEI @100 x 1C | ≤ 500 nm | not reached (baseline Chen2020 ~449 nm signal) | n/a |

| True-voltage guard | ≥ 4.6 beyond proxy | envelope re-computation 4.6354 V (independent run) | informative |

Overall: NOT ACHIEVED. The only window-passer fails the electrolyte compatibility limit by

2.6 V; no candidate passes both stage-1 co-gates, so stages 2-4 were never reached.